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SECOND SEMESTER M.TECH EXAMINATION, JUNE – 2019

DAIRY TECHNOLOGY

Time: 3 Hours **Maximum Marks: 100**

Instructions: **i. Answer FIVE full questions. Question No. 1 and 2 are compulsory.**
ii. Draw neat sketches wherever necessary.
iii. Make suitable assumptions wherever necessary.

1.	**a.**	Among the various physico-chemical properties of milk, discuss in detail on density and specific gravity. Express the formula used to calculate percentage TS and percentage SNF.	**10**
	b.	Recall the minor milk constituents and their significance.	**10**
2.	**a.**	Summarize the flavor defects in milk, their causes and prevention with examples.	**10**
	b.	Substantiate on "Selection and precautions in the use of detergents and sanitizers for different surface materials of dairy equipments".	**10**
3.	**a.**	With the help of a neat flow diagram, illustrate pasteurization process.	**08**
	b.	Paraphrase the process involved in production of homogenized milk.	**12**

OR

4	**a.**	Examine the factors influencing homogenization. Extend a note on the merits and demerits of homogenization.	**10**
	b.	Illustrate the packaging systems used for packing the pasteurized milk.	**10**
5.	**a.**	Inspect the various types of cream and the per cent of fat in each type of cream.	**10**
	b.	"Butter making is an inversion of the original cream emulsion". Critically comment on the statement.	**10**

OR

6.	**a.**	Elaborate on frozen cream with a special emphasis on the effects of freezing.	**10**
	b.	Summarize the methods for judging and grading of table butter.	**10**

7. Narrate the applications of following enzymes in dairy industry: **20**

i. Proteases
ii. Lipases
iii. Esterases
iv. Lactase

OR

8. Prioritize the applications of the following in dairy industry **i.** microbial rennet in dairy applications **ii.** Microbial enzymes in accelerated cheese ripening **iii.** lactase treated milk and milk products **iv.** enzyme modified cheese **20**

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